

* Rank 0 tensor. * Rank 1 tensor. * Rank 2 tensor (s). * Rank 2 tensor (a).

$$\begin{bmatrix} 0 & M_{xy} & 0 \\ -M_{xy} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{xy} = \mathbb{T}_z^{(1)}$$

* Rank 3 tensor (ss). * Rank 3 tensor (sa). * Rank 3 tensor (a). * Rank 4 tensor (sss).

$$\begin{bmatrix} 0 & 0 & 0 & M_{xxyz} & 0 & 0 \\ 0 & 0 & 0 & -M_{xxyz} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ M_{xxyz} & -M_{xxyz} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & M_{xxyz} \\ 0 & 0 & 0 & 0 & M_{xxyz} & 0 \end{bmatrix}$$

$$M_{xxyz} = \mathbb{M}_{4a}^{(1)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & 0 & M_{xxyz} & 0 & M_{xxxy} \\ 0 & 0 & 0 & -M_{xxyz} & 0 & -M_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -M_{xxyz} & M_{xxyz} & 0 & 0 & M_{yzzx} & 0 \\ 0 & 0 & 0 & -M_{yzzx} & 0 & -M_{xxyz} \\ -M_{xxxy} & M_{xxxy} & 0 & 0 & M_{xxyz} & 0 \end{bmatrix}$$

$$M_{xxyz} = 2\mathbb{T}_{3a}^{(1)}$$

$$M_{xxxy} = 2\mathbb{T}_z^{(1)} - 2\mathbb{T}_z^{\alpha(1)}$$

$$M_{yzzx} = -\mathbb{T}_z^{(1)} - 4\mathbb{T}_z^{\alpha(1)}$$

* Rank 4 tensor (aas). * Rank 4 tensor (aaa).

$$\begin{bmatrix} 0 & M_{yzzx} & 0 \\ -M_{yzzx} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzzx} = -\mathbb{T}_z^{(2)}$$

* Rank 4 tensor (sa).

$$\begin{bmatrix} M_{xxyz} & 0 & M_{xxxy} \\ -M_{xxyz} & 0 & M_{xxxy} \\ 0 & 0 & M_{zzxy} \\ 0 & M_{yzzx} & 0 \\ M_{yzzx} & 0 & 0 \\ 0 & -M_{xxyz} & 0 \end{bmatrix}$$

$$M_{xxyz} = \mathbb{T}_{3a}^{(2)}$$

$$M_{xxxy} = \mathbb{T}_z^{(3)} - \mathbb{T}_z^{\alpha(2)}$$

$$M_{zzxy} = \mathbb{T}_z^{(3)} + 2\mathbb{T}_z^{(4)} + 2\mathbb{T}_z^{\alpha(2)}$$

$$M_{yzzx} = \mathbb{T}_z^{(4)} - \mathbb{T}_z^{\alpha(2)}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} M_{yzxx} & -M_{yzxx} & 0 & 0 & M_{yzzx} & 0 \\ 0 & 0 & 0 & M_{yzzx} & 0 & -M_{yzxx} \\ M_{xyxx} & M_{xyxx} & M_{xyzx} & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{aligned} M_{yzxx} &= \mathbb{T}_{3a}^{(3)} \\ M_{yzzx} &= \mathbb{T}_z^{(6)} - \mathbb{T}_z^{\alpha(3)} \\ M_{xyxx} &= \mathbb{T}_z^{(5)} - \mathbb{T}_z^{\alpha(3)} \\ M_{xyzx} &= \mathbb{T}_z^{(5)} + 2\mathbb{T}_z^{(6)} + 2\mathbb{T}_z^{\alpha(3)} \end{aligned}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} 0 & 0 & 0 & M_{xxyz} & 0 & M_{xxxy} & M_{xxzy} & 0 & M_{xxyx} \\ 0 & 0 & 0 & -M_{xxyz} & 0 & -M_{xxyx} & -M_{xxzy} & 0 & -M_{xxxy} \\ 0 & 0 & 0 & 0 & 0 & M_{zzxy} & 0 & 0 & -M_{zzxy} \\ M_{yzxx} & -M_{yzxx} & 0 & 0 & M_{yzzx} & 0 & 0 & M_{yzxz} & 0 \\ 0 & 0 & 0 & -M_{yzxz} & 0 & M_{yzxx} & -M_{yzxz} & 0 & M_{yzxx} \\ -\frac{M_{xxxy}}{2} - \frac{M_{xxzy}}{2} & \frac{M_{xxxy}}{2} + \frac{M_{xxzy}}{2} & 0 & 0 & M_{xxzy} & 0 & 0 & M_{xxzy} & 0 \end{bmatrix}$$

$$\begin{aligned} M_{xxyz} &= \mathbb{M}_{4a}^{(1)} + 2\mathbb{T}_{3a}^{(1)} + \mathbb{T}_{3a}^{(2)} \\ M_{xxxy} &= 2\mathbb{T}_z^{(1)} + \mathbb{T}_z^{(3)} - 2\mathbb{T}_z^{\alpha(1)} - \mathbb{T}_z^{\alpha(2)} \\ M_{xxzy} &= \mathbb{M}_{4a}^{(1)} + 2\mathbb{T}_{3a}^{(1)} - \mathbb{T}_{3a}^{(2)} \\ M_{xxyx} &= 2\mathbb{T}_z^{(1)} - \mathbb{T}_z^{(3)} - 2\mathbb{T}_z^{\alpha(1)} + \mathbb{T}_z^{\alpha(2)} \\ M_{zzxy} &= \mathbb{T}_z^{(3)} + 2\mathbb{T}_z^{(4)} + 2\mathbb{T}_z^{\alpha(2)} \\ M_{yzxx} &= \mathbb{M}_{4a}^{(1)} - 2\mathbb{T}_{3a}^{(1)} \\ M_{yzzx} &= -\mathbb{T}_z^{(1)} + \mathbb{T}_z^{(4)} - 4\mathbb{T}_z^{\alpha(1)} - \mathbb{T}_z^{\alpha(2)} \\ M_{yzxz} &= -\mathbb{T}_z^{(1)} - \mathbb{T}_z^{(4)} - 4\mathbb{T}_z^{\alpha(1)} + \mathbb{T}_z^{\alpha(2)} \end{aligned}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} M_{yzxx} & -M_{yzxx} & 0 & 0 & M_{yzzx} & 0 & 0 & M_{yzxz} & 0 \\ 0 & 0 & 0 & M_{yzxz} & 0 & -M_{yzxx} & M_{yzzx} & 0 & -M_{yzxx} \\ M_{xyxx} & M_{xyxx} & M_{xyzx} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{aligned} M_{yzxx} &= \mathbb{T}_{3a}^{(3)} \\ M_{yzzx} &= -\mathbb{T}_z^{(2)} + \mathbb{T}_z^{(6)} - \mathbb{T}_z^{\alpha(3)} \\ M_{yzxz} &= \mathbb{T}_z^{(2)} + \mathbb{T}_z^{(6)} - \mathbb{T}_z^{\alpha(3)} \\ M_{xyxx} &= \mathbb{T}_z^{(5)} - \mathbb{T}_z^{\alpha(3)} \\ M_{xyzx} &= \mathbb{T}_z^{(5)} + 2\mathbb{T}_z^{(6)} + 2\mathbb{T}_z^{\alpha(3)} \end{aligned}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & M_{yyyy} \\ 0 & 0 & 0 \\ 0 & M_{yyyy} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -M_{yyyy} \\ 0 & 0 & 0 \\ 0 & -M_{yyyy} & 0 \\ 0 & 0 & 0 \\ -M_{yyyy} & 0 & 0 \end{bmatrix}$$

$$M_{yyyy} = -\mathbb{M}_{4a}^{(1)}$$